1.4 NULL HYPOTHESIS AND ALTERNATIVE HYPOTHESIS

In many practical studies, as mentioned earlier, it is necessary to make decisions about a population or its unknown characteristics on the basis of sample observations. For example, in bio-medical studies, we may be investigating a particular theory that the recently developed medicine is much better than the conventional medicine in curing a disease. For this purpose, we propose a statement on the population or the theory. Such statements are called hypotheses.

Thus, a **hypothesis** can be defined as a statement on the population or the values of the unknown parameters associated with the respective probability distribution. All the hypotheses should be tested for their validity using statistical concepts and a representative sample drawn from the study population. '*Hypotheses*' is the plural form of '*hypothesis*'.

A **statistical test** is a procedure governed by certain determined/derived rules, which lead to take a decision about the null hypothesis for its rejection or otherwise on the basis of sample values. This process is called **statistical hypotheses testing**.

The statistical hypotheses testing plays an important role, among others, in various fields including industry, biological sciences, behavioral sciences and Economics. In each hypotheses testing problem, we will often find as there are two hypotheses to choose between *viz*., null hypothesis and alternative hypothesis.

Null Hypothesis:

A hypothesis which is to be actually tested ***for possible rejection*** based on a random sample is termed as **null hypothesis**, which will be denoted by H_0.

YOU WILL KNOW

(i) Generally, it is a hypothesis of no difference in the case of comparison.

(ii) Assigning a value to the unknown parameter in the case of single sample problems

(iii) Suggesting a suitable model to the given environment in the case of model construction.

(iv) The given two attributes are independent in the case of *Chi*-square test for independence of attributes.

Alternative Hypothesis:

A statement about the population, which contradicts the null hypothesis, depending upon the situation, is called **alternative hypothesis**, which will be denoted by H_1.

For example, if we test whether the population mean has a specified value μ_0, then the null hypothesis would be expressed as:

$$H_0: \mu = \mu_0$$

The alternative hypothesis may be formulated suitably as anyone of the following:

(i) $H_1: \mu \neq \mu_0$

(ii) $H_1: \mu > \mu_0$

(iii) $H_1: \mu < \mu_0$

12th_Statistics_EM_Unit_1.indd 7 1/28/2020 4:47:29 PM